

Big Data Analytics for Multimodal Misinformation Detection using Large Language Models (LLMs) and Vision Language Models (VLMs)

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1. Introduction

Objective: To develop a scalable framework that leverages the reasoning capabilities of Large Language Models (LLMs) and Vision Language Models (VLMs) to detect multimodal misinformation on social media.

Problem Statement: The proliferation of fake news on platforms like Weibo involves complex interactions between text and images. Traditional unimodal models struggle with "semantic gaps" where the text and image appear authentic individually but contradict each other contextually.

2. Literature Review

Foundational Methods: Early approaches relied on handcrafted features or simple co-attention fusion mechanisms such as MCAN. **Modern LLM/VLM Integration:** Recent shifts utilize Retrieval-Augmented Generation (RAG) to ground LLM predictions in external, real-time evidence, reducing the risk of "hallucinations". **Research Gaps:** Cross-Domain Generalization: Most models perform well on specific datasets but fail when transferred to new news domains. **Generative AI Challenges:** The rise of GenAI-driven "news diversity" (highly varied synthetic content) has introduced "model-level misperception drift," significantly degrading the robustness of 2026 detectors.

3. Methodology

Dataset: The Weibo dataset, a benchmark for Chinese social media misinformation containing multimodal posts (text and associated images).

Proposed Model Architecture: **Text Processing:** Utilizing BERT or similar LLMs for deep semantic extraction. **Visual Analysis:** Implementing Vision Transformers (ViT) or CLIP-based encoders to extract visual features. **Multimodal Fusion:** A cross-modal consistency check that aligns textual and visual embeddings to detect subtle mismatches.

Experimental Setup: The system is built using Python, PyTorch, and HuggingFace Transformers, designed for scalability on big data platforms.

4. Results

Metrics: Include a table comparing Accuracy, F1-Score, Precision, and Recall. SOTA models in 2026 (like M&M-RAG) have achieved up to 94.6% accuracy on similar multimodal benchmarks. **Analysis:** Provide a confusion matrix showing the model's ability to distinguish "Non-Rumor" from "Rumor". **Ablation Study:** Demonstrate the necessity of multimodal fusion by showing decreased performance when using text or images alone.

5. Conclusion

Developed a unified framework for detecting Weibo rumors by combining LLM reasoning with visual forensic analysis. **Future Work:** Integrating Visual Forgery Detection specifically for AI-generated images. Expanding the system to support low-resource languages beyond English and Chinese.

6. References

- [1] Li, X., et al. (2025). A Survey of Multimodal Fake News Detection: A Cross-Modal Interaction Perspective. IEEE Transactions on Emerging Topics in Computational Intelligence.
- [2] Harris, E., et al. (2026). Multimodal misinformation detection across diverse languages using RAG and LLMs. Knowledge Management & E-Learning.